

Customer Shopping Behavior Exploratory Data Analysis (EDA) Report

A complete analysis of purchasing patterns, customer preferences, and business recommendations

Project Type: EDA | Contribution: Individual

Tools: Python (Pandas, Matplotlib, Seaborn)

Project Summary

This project focuses on analyzing customer shopping behavior for a retail company using Exploratory Data Analysis (EDA). The objective is to identify purchasing patterns, customer preferences, and the factors that influence buying decisions. By examining customer demographics, product categories, shopping frequency, payment methods, seasonal trends, discounts, and purchase amounts, the analysis aims to generate actionable business insights that can support strategic decision-making.

The project was carried out using Python in Google Colab with the Pandas, Matplotlib, and Seaborn libraries. The dataset was explored using statistical techniques such as `.info()`, `.describe()`, `.value_counts()`, and missing value analysis to understand its structure and ensure data quality. Various visualizations, including histograms, boxplots, count plots, bar plots, a pairplot, and a correlation heatmap, were created to identify trends, distributions, relationships, and potential anomalies within the data.

These findings enable the retail company to better understand its customers and make informed business decisions — improving inventory management, optimizing promotional campaigns, personalizing customer experiences, enhancing pricing strategies, and strengthening customer retention.

Problem Statement

A leading retail company aims to gain a deeper understanding of customer shopping behavior to improve business performance, customer satisfaction, and long-term customer loyalty. With changing consumer preferences and increasing competition in the retail industry, the company has observed noticeable variations in purchasing patterns across different customer demographics, product categories, and shopping channels.

Management wants to identify the key factors that influence customers' purchasing decisions — including age, gender, product preferences, seasonal trends, discounts, payment methods, customer reviews, and shopping frequency — and how these factors affect purchase amounts and repeat buying behavior.

Business Objective

The primary objective of this project is to analyze customer shopping behavior and identify the key factors that influence purchasing decisions. Using Exploratory Data Analysis, the project aims to uncover patterns related to customer demographics, product categories, purchase amounts, shopping frequency, payment methods, discounts, and seasonal trends — enabling the company to optimize marketing strategies, improve inventory planning, personalize customer experiences, and develop targeted promotional campaigns.

Dataset Overview

The dataset consists of 3,900 customer records and 18 columns, containing customer demographics, purchase details, product information, and payment methods. Only the Review Rating column had 37 missing values (0.95%), while all other columns were complete. No duplicate records were found, indicating the dataset was clean and ready for analysis.

Data Wrangling

A copy of the original dataset (`df_copy`) was created to preserve the raw data. Duplicate records were checked using `duplicated().sum()`, and 0 duplicates were found, confirming all customer records are unique. The 37 missing values in the Review Rating column were imputed with the median value to preserve the data distribution and avoid the influence of outliers. No rows were removed during cleaning, so the final dataset retained its full shape of 3,900 rows by 18 columns — clean and ready for Exploratory Data Analysis.

Chart-wise Findings

Chart 1: Age Distribution

Why this chart: A histogram with KDE is the most suitable chart for visualizing the distribution of a continuous numerical variable like Age, revealing how customer ages are spread and the frequency of customers in each age range.

Insight: The majority of customers belong to the middle-age group, with comparatively fewer customers in the younger and older age groups. The distribution is fairly balanced without extreme concentration in a single age range, indicating the business attracts customers across a wide age spectrum, with the highest participation from middle-aged customers.

Business impact: Understanding age distribution helps design age-specific marketing campaigns, recommend products by age group, and improve inventory planning. There's no direct negative signal, but if one age group dominates while others are underrepresented, the business may be missing growth opportunities in those segments.

Chart 2: Gender Distribution

Why this chart: A count plot is suitable for comparing the number of customers in each gender category.

Insight: The dataset contains significantly more male customers (~2,650) than female customers (~1,250) — male customers form the clear majority of the customer base.

Business impact: Gender-specific marketing campaigns and personalized offers can improve retention among the dominant male segment. However, the comparatively smaller female segment represents an untapped growth opportunity — continuing to focus mainly on male customers risks missing sales growth from female shoppers, so targeted campaigns aimed at this group could help balance customer growth and increase overall revenue.

Chart 3: Category Distribution

Why this chart: A count plot compares purchase frequency across product categories.

Insight: The chart reveals the most and least purchased categories, highlighting clear demand differences across the product range.

Business impact: Popular categories should receive higher inventory and promotional focus to capture existing demand.

Chart 4: Purchase Amount Distribution

Why this chart: A histogram with KDE overlay shows the overall shape and spread of a continuous variable like purchase amount, including whether it's skewed, uniform, or multi-peaked.

Insight: Purchase amounts are fairly evenly spread across the range with no extreme skew and no dominant single spending tier — customers spend broadly similar amounts rather than clustering at one price point.

Business impact: No negative signal; a flat distribution is an opportunity to design targeted upsell bundles or premium tiers to lift the lower-spending segment.

Chart 5: Purchase Amount by Gender

Why this chart: A boxplot compares median, spread, and outliers of purchase amount across gender categories in one view.

Insight: Median purchase amount and spread are quite similar across genders, with comparable outlier ranges — spending behavior isn't strongly differentiated by gender.

Business impact: Gender-based pricing or discounting isn't necessary; marketing spend is better directed at other differentiators such as category or season. No negative impact identified.

Chart 6: Average Purchase Amount by Category

Why this chart: A bar plot is ideal for comparing an average metric (purchase amount) across a categorical variable (product category).

Insight: Average spend varies by category, with some categories commanding a noticeably higher average purchase amount than others.

Business impact: High-average-spend categories are strong candidates for premium marketing and cross-selling; lower-average categories can be bundled with higher-margin items. No negative growth risk — this simply guides where to focus revenue-boosting efforts.

Chart 7: Season-wise Purchases

Why this chart: A count plot clearly shows how purchase volume is distributed across the four seasons.

Insight: Purchase counts are relatively balanced across seasons, though one or two seasons show slightly higher activity, hinting at mild seasonality in buying behavior.

Business impact: Knowing peak seasons helps with inventory planning and timing promotions; a season with notably lower purchases could signal the need for a targeted seasonal campaign to avoid a slow-sales period.

Chart 8: Payment Method

Why this chart: A horizontal count plot works well for a categorical variable with several method names, making labels easy to read.

Insight: Certain payment methods (e.g., card-based options) are used far more frequently than others, showing a clear customer preference.

Business impact: Priority can go to partnerships, fee negotiations, or UX improvements for the most-used methods; rarely used methods could be phased out to reduce checkout complexity — an efficiency opportunity, not a negative signal.

Chart 9: Subscription Status

Why this chart: A simple count plot best shows the split between subscribed and non-subscribed customers.

Insight: A noticeably smaller proportion of customers are subscribed compared to non-subscribers, indicating subscription adoption is currently low.

Business impact: Clear growth lever: low subscription uptake represents a real opportunity to boost recurring revenue with a stronger subscription incentive or loyalty program. Left unaddressed, this represents missed recurring-revenue potential.

Chart 10: Shipping Type

Why this chart: A horizontal count plot suits comparing frequency across multiple shipping type labels.

Insight: One or two shipping types dominate customer choice, while others are used far less often.

Business impact: The business can optimize logistics costs and delivery partnerships around the most-used shipping options; underused types could be promoted (if cheaper) or discontinued to simplify operations.

Chart 11: Review Rating Distribution

Why this chart: A histogram with KDE shows the shape of the review rating distribution and whether ratings skew positive or negative.

Insight: Ratings are concentrated in the mid-to-upper range, suggesting most customers are moderately to highly satisfied, with relatively few very low ratings.

Business impact: Positive ratings support satisfaction-based marketing. If a meaningful tail of low ratings exists, that segment should be investigated further (e.g., which categories or shipping types they correlate with) to prevent churn — a potential negative-growth signal if ignored.

Chart 12: Discount Applied

Why this chart: A count plot clearly shows how many purchases involved a discount versus none.

Insight: A substantial share of purchases are made with a discount applied, suggesting price sensitivity plays a real role in purchase decisions.

Business impact: Discounts clearly drive purchase volume, but heavy reliance on discounting can erode margins over time — a potential negative-growth risk if not balanced with full-price sales strategies or loyalty-based (rather than blanket) discounting.

Chart 13: Promo Code Usage

Why this chart: A count plot is the simplest way to show the binary split between promo-code users and non-users.

Insight: A significant portion of customers use promo codes, closely mirroring the discount-usage pattern, reinforcing that promotions are a strong purchase driver.

Business impact: Largely positive — promo codes are effective for targeted campaigns (e.g., win-back offers). The same margin-erosion caution as discounts applies if promo usage becomes the default rather than the exception.

Chart 14: Correlation Heatmap

Why this chart: A heatmap shows relationships among all numerical variables at once.

Insight: Most numerical variables show very weak or no correlation. Age and Previous Purchases have a very weak positive correlation (approx. 0.04), indicating age has little influence on previous purchase frequency. Purchase Amount and Review Rating have a very weak positive correlation (approx. 0.03), suggesting spending slightly increases with higher ratings, though the relationship is negligible. Customer ID shows no meaningful relationship with any variable, as expected since it is only a unique identifier. Overall, no strong positive or negative correlations exist among the numerical variables.

Business impact: No strong linear relationships exist among numerical variables, so predictive modeling should lean on categorical features and interactions rather than simple numeric correlations.

Chart 15: Pair Plot

Why this chart: A pair plot analyzes relationships among multiple numerical variables (Age, Purchase Amount, Review Rating, Previous Purchases) simultaneously.

Insight: The plot shows trends, distributions, and possible outliers across variable pairs, useful for spotting patterns not visible in a single correlation number.

Business impact: Useful foundation for customer segmentation and predictive modeling efforts.

Solution to Business Objective

What should the client do to achieve the business objective?

To help the client grow revenue, improve customer retention, and operate more efficiently, the following actions are recommended based on the EDA findings:

1. **Focus inventory and promotion on top-performing categories.** Categories with higher purchase frequency and higher average purchase amount (Charts 3 and 6) should get priority shelf space, ad spend, and stock levels, since they already have proven demand.
2. **Invest in growing the subscription base.** Subscription adoption is currently low (Chart 9). Introduce stronger incentives — free shipping, early access to sales, or subscriber-only discounts — to convert one-time buyers into recurring subscribers, which stabilizes revenue over time.
3. **Use discounts and promo codes strategically, not as a default.** A large share of purchases already involve a discount or promo code (Charts 12 and 13). Rather than broad, always-on discounting (which erodes margin), shift toward targeted promotions — e.g., win-back offers for lapsed customers or first-purchase incentives — while encouraging full-price purchases from loyal customers through non-discount perks.
4. **Streamline payment and shipping options.** Since a small number of payment methods and shipping types account for most usage (Charts 8 and 10), simplify checkout by promoting these preferred options, potentially reducing operational costs from maintaining rarely-used alternatives.
5. **Plan around seasonal demand.** Purchases show mild seasonal variation (Chart 7). Align inventory build-up and marketing campaigns with the higher-demand season(s), and design targeted promotions to lift sales in the slower season.
6. **Monitor and act on review ratings.** While most ratings are moderate-to-high (Chart 11), set up a process to flag and follow up on low-rated purchases quickly (e.g., customer service outreach), since unaddressed dissatisfaction is a direct churn risk.
7. **Use purchase amount and previous-purchase patterns for segmentation.** Since numerical variables show weak correlation with each other (Chart 14) but purchase amount, review rating, and previous purchases together reveal customer behavior patterns (Chart 15), build simple customer segments (e.g., high-value/low-frequency vs. low-value/high-frequency) to personalize marketing rather than treating all customers the same.

Together, these steps target the two levers that move the business objective forward: **increasing revenue per customer** (via category focus, smarter discounting, and segmentation) and **increasing customer retention** (via subscriptions, seasonal planning, and proactive service on low ratings).

Conclusion

This EDA capstone project analyzed customer shopping behavior across demographics, product categories, payment and shipping preferences, discounts, and review ratings. The data showed weak correlations among numerical variables, fairly balanced spending across genders and seasons, and clear customer preferences in payment method and shipping type. Key opportunities emerged around growing subscription adoption, managing discount and promo reliance to protect margins, and using category- and season-level insights to guide inventory and marketing decisions. Overall, the analysis provides a solid foundation for data-driven decisions to improve customer engagement and drive sustainable business growth.